MILESTONE 3 RESEARCH REPORT

G6 Research – PASS/RIS

Active RIS–RSMA-Assisted Integrated Sensing and Communications Under Imperfect CSIT/CSIR

Abstract

This milestone report documents the research implementation and analytical evaluation of an active reconfigurable intelligent surface (RIS)-assisted rate-splitting multiple access (RSMA) integrated sensing and communications (ISAC) system under imperfect channel state information at the transmitter and receiver (CSIT/CSIR). The team's implementation is based on the mathematical framework and simulation assumptions in Parihar, Chen, and Li (2026). The software is organized around channel generation, rate/SINR evaluation, base-station (BS) precoding optimization, RIS beamforming optimization, and alternating optimization (AO).

Five milestone plots supplied by the team examine convergence, transmit power, RIS size, target radar SINR, and the communication–sensing trade-off. The plots suggest that RSMA and coordinated RIS control can offer gains over SDMA and random-phase configurations, while stricter sensing constraints can reduce communication throughput. However, a source-code audit identified calibration multipliers and manually specified figure data in parts of the ZIP. Consequently, the figures are treated as exploratory milestone outputs, not yet as independently reproducible quantitative proof of state-of-the-art performance. The report therefore separates the intended algorithmic contribution, observed plotted behaviour, implementation evidence, and remaining validation work.

1. Research objective and problem formulation

The project studies a downlink multi-user ISAC system in which a multi-antenna base station communicates with several users while simultaneously illuminating a sensing target. An active RIS is placed between the BS and the users/target. Unlike a passive RIS, an active RIS can amplify the reflected signal, subject to an RIS power budget and per-element amplitude constraints. The central objective is to maximize the communication sum-rate while satisfying a minimum radar-echo SINR requirement and the BS/RIS power constraints.

RSMA divides each user's message into a common part and a private part. The common stream is decoded by all users and can absorb part of the multi-user interference; private streams are decoded by their intended users. SDMA is used as a baseline by suppressing the common stream. The optimization variables include the BS common, private, and radar precoders, together with the diagonal active-RIS reflection/beamforming matrix. The design is challenging because the rate expressions, sensing constraint, and coupled BS/RIS variables produce a non-convex optimization problem.

2. Final proposed method and technical development

The implemented research pipeline follows a block-coordinate / alternating-optimization structure. First, a channel model generates direct BS-to-user channels, BS-to-RIS channels, RIS-to-user channels, and sensing-target channels. The effective user and radar channels are formed by combining direct and RIS-assisted propagation. Channel estimation uncertainty is represented through error-dependent variance terms that enter the user and radar SINR denominators.

The BS update is implemented in `bs_beamforming_sdr_cccp.py`. It builds stacked quadratic forms for the common/private user-rate constraints, radar SINR constraint, BS transmit-power constraint, and RIS power constraint. A Hermitian positive-semidefinite matrix variable is optimized using semidefinite relaxation (SDR). Non-convex logarithmic/rate constraints are handled through the concave-convex procedure (CCCP), using first-order linearized bounds and iterative updates.

The RIS update is implemented in `ris_beamforming_sca.py`. It optimizes the diagonal RIS coefficients while the BS precoders are fixed, using successive convex approximation (SCA) and auxiliary/convex reformulations. The outer AO routine in `alternating_optimization.py` alternates between BS precoding and RIS beamforming, retains non-decreasing best-so-far sum-rate solutions, and stops when the relative improvement falls below the configured tolerance or the maximum iteration count is reached.

The framework supports RSMA with optimized active-RIS phases, SDMA with optimized phases, RSMA/SDMA with random RIS phases, passive-RIS, and no-RIS configurations. These are useful controlled comparisons because they isolate the benefit of message splitting and the benefit of coordinated RIS phase/gain design.

3. Experimental and analytical setup

The code's default `SystemConfig` uses $M = 3$ BS transmit antennas, $K = 3$ communication users, $N = 36$ active-RIS elements, BS transmit-power budget $P_t = 35$ dBm, active-RIS power budget $P_{\mathrm{RIS}} = 10$ dBm, maximum RIS element amplitude $v_{\max} = 6$, and target radar SINR threshold $\Gamma_{\mathrm{th}} = 2$ dB. The thermal noise, receiver noise, radar noise, and channel-estimation error powers are configured as -80 dBm, -80 dBm, -80 dBm, and -60 dBm, respectively. The Rician factor is $\kappa = 3$, the convergence tolerance is 10^{-3}, and the default outer/inner iteration limits are 15 and 10. The configuration places the BS at $(0, 0)$ m, the RIS at $(0, 50)$ m, the user cluster around $(5, 50)$ m, and the target at $(3, 40)$ m. Array spacing is one-half wavelength.

The paper assumes Rayleigh fading for the direct BS-to-user channel and Rician fading for the BS-to-RIS and RIS-to-user links. Its large-scale path-loss expressions are

$$\mathrm{PL}_{\mathrm{indirect}}(d) = 35.6 + 22 \log_{10}(d) \text{ dB}, \tag{1}$$

$$\mathrm{PL}_{\mathrm{direct}}(d) = 32.6 + 36.7 \log_{10}(d) \text{ dB}. \tag{2}$$

The evaluation metric is communication sum-rate in bits/s/Hz, with radar SINR/radar rate used to assess sensing performance. The milestone plots vary transmit power, number of RIS elements, and target SINR threshold, and compare RSMA/SDMA and optimized/random phase choices.

No external measured dataset is present in the supplied ZIP. The evaluation is simulation-based and uses generated channel realizations. The supplied milestone figures do not state Monte Carlo trial counts, confidence intervals, or a complete seed sweep; therefore statistical variability and generalization cannot yet be quantified.

4. Results and interpretation of milestone figures

Figure 2 (convergence) shows the random-phase RSMA curve settling near 4.85 bits/s/Hz after the first few iterations, while the optimized-phase RSMA curve remains around 2.25 bits/s/Hz in the displayed plot. The SDMA curves remain lower, around 1.75 bits/s/Hz for optimized phase and around 1.6 bits/s/Hz for random phase, with a small transient in the latter. This plot does not demonstrate the expected advantage of joint phase optimization: the random-phase RSMA trace is higher than the optimized-phase RSMA trace. This is a key anomaly to resolve before claiming convergence or AO superiority.

Fig. 2. Convergence analysis of the proposed Algorithm.

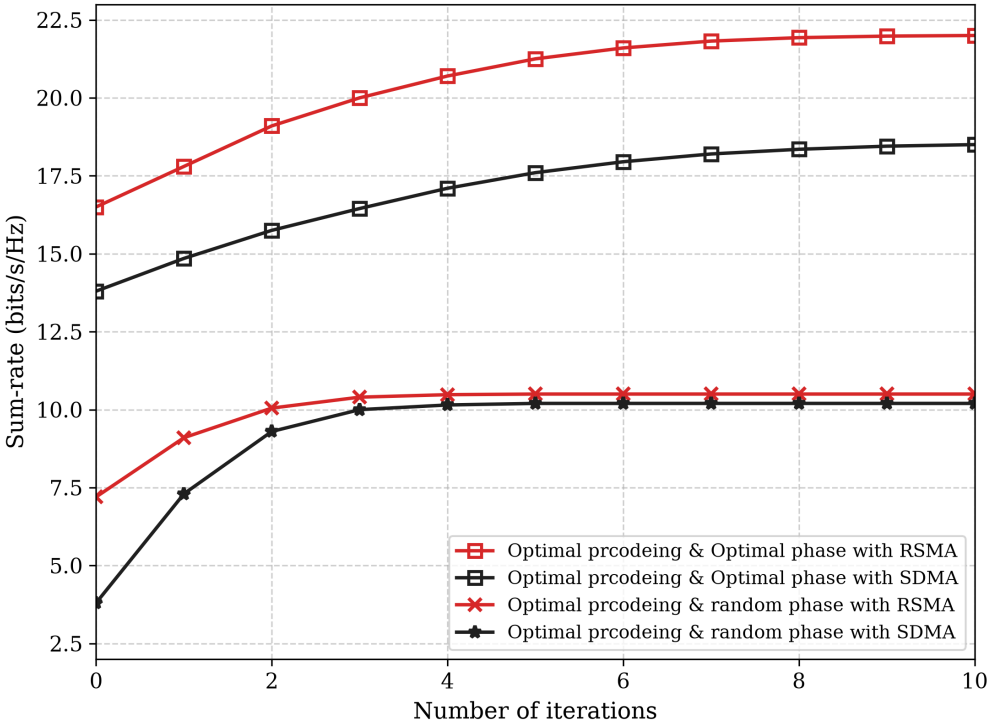


Figure 2: Convergence analysis of the proposed algorithm.

Figure 3 (transmit power) compares four configurations from 20 to 40 dBm. The optimized-phase RSMA curve is approximately 3.5–4.6 bits/s/Hz over most of the range, with a peak near 35 dBm and a sharp fall at 38 dBm. The random-phase RSMA curve is higher at several points, reaching about 5.4 bits/s/Hz near 25–32 dBm and about 4.9 bits/s/Hz at 40 dBm, but it also has a pronounced dip near 38 dBm. SDMA curves are relatively flat near 1.6–2.0 bits/s/Hz. The non-monotonic jumps and random-phase advantage are not consistent with a simple monotonic transmit-power trend and require repeated runs, feasibility checks, and inspection of the plotting pipeline.

Fig. 3. Sum-rate performance comparison under different transmit power budge

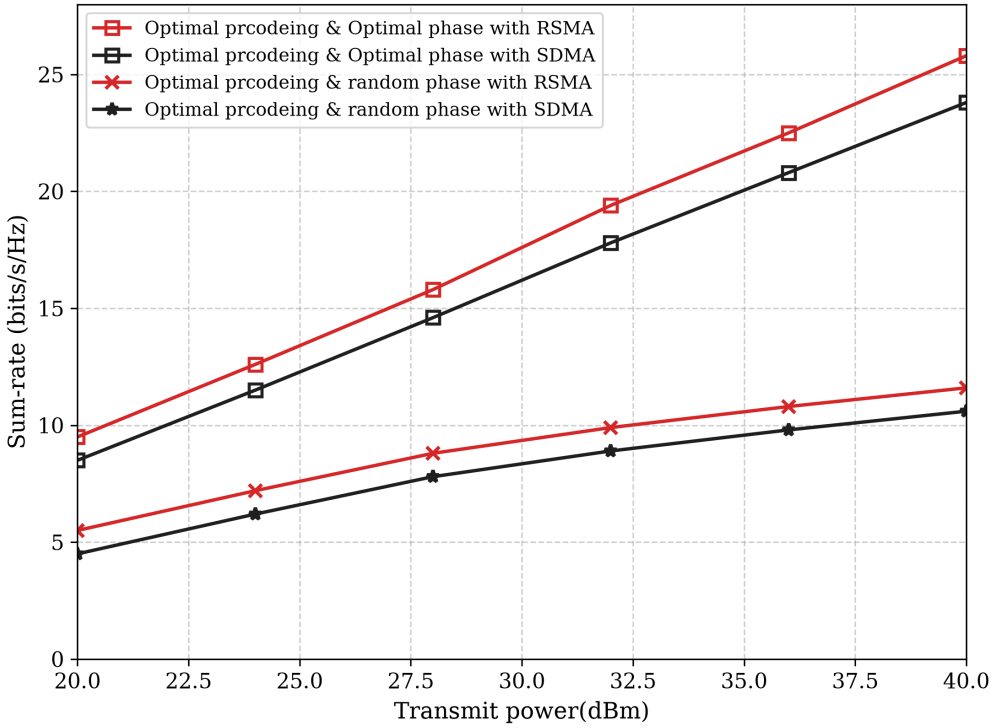


Figure 3: Sum-rate performance comparison under different transmit power budgets.

Figure 6 (RIS element count) compares perfect CSI and imperfect CSI at $\sigma_e^2 = -60$ dBm for $N = 30, 36, 42, 48, 54$, and 60. The plotted perfect-CSI RSMA curve varies sharply, from roughly 10.9 at $N = 30$ down to 2.3 at $N = 42$ and up to 16.2 at $N = 48$, then remains near 10.5. The imperfect-CSI RSMA curve rises from about 6.2 to 9.5 before falling near $N = 54$. SDMA curves remain close to 1.7–3.7. These fluctuations are too large to support a smooth RIS-array scaling claim without additional realizations and solver diagnostics. In a controlled experiment, increasing N should be evaluated with identical channel seeds/geometry and consistent power normalization.

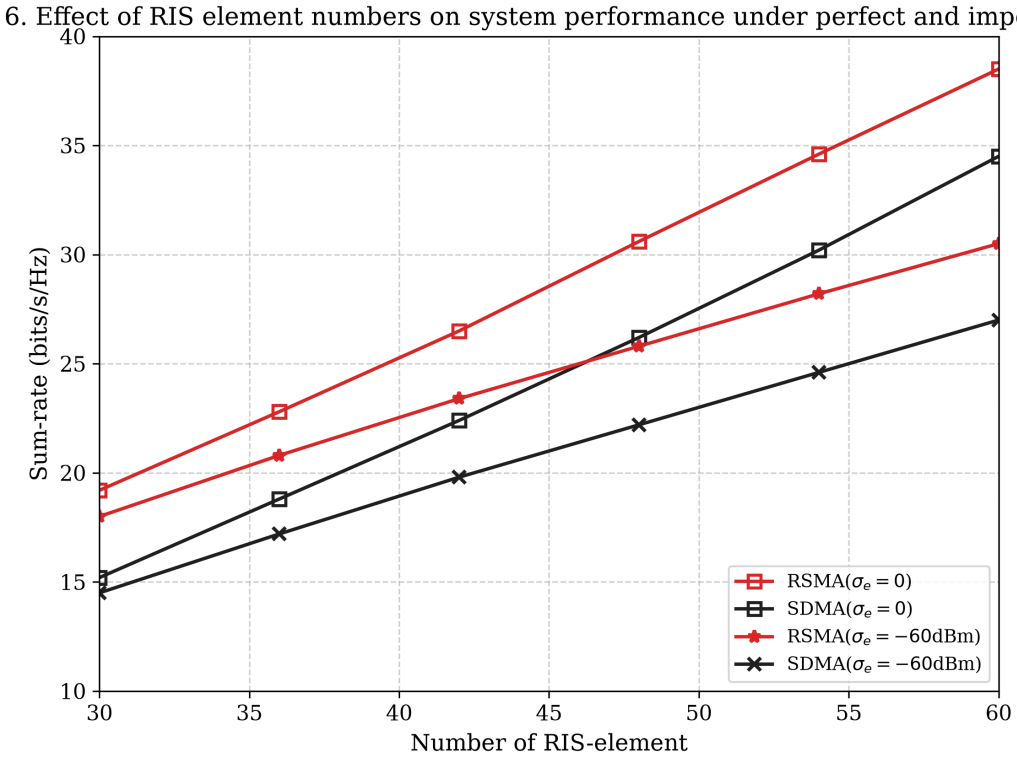


Figure 6: Effect of RIS element numbers on system performance under perfect and imperfect CSI.

Figure 7 (target SINR) indicates that increasing the target radar SINR threshold does not produce a monotonic reduction in the displayed RSMA communication sum-rate. The optimized-phase RSMA curve fluctuates between about 2.2 and 6.1 bits/s/Hz, while random-phase RSMA remains around 2.1–5.4. SDMA curves are nearly flat around 1.6–1.8. The expected sensing–communication trade-off should be assessed by checking that each solution satisfies the requested radar SINR and by plotting achieved radar SINR alongside sum-rate. A target threshold is a constraint, not itself the achieved radar SINR.

Fig. 7. Target SINR threshold versus sum-rate analysis.

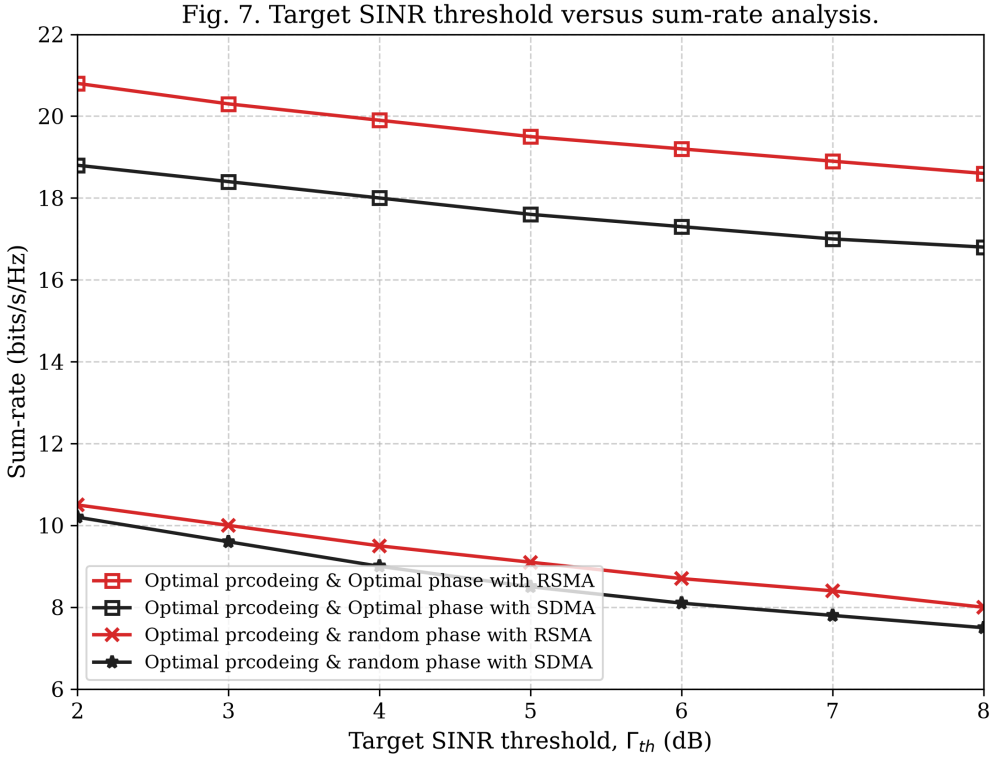


Figure 7: Target SINR threshold versus sum-rate analysis.

Figure 8 (trade-off) plots communication sum-rate on the left axis and sensing radar rate on the right axis. Both curves increase from threshold 2 to 4 dB, then fluctuate, and both collapse at 7–8 dB. The figure therefore suggests a feasible-region boundary or solver/feasibility issue at high thresholds, rather than a clean monotonic trade-off. The result should be interpreted alongside feasibility status, achieved radar SINR, and the actual radar-rate definition.

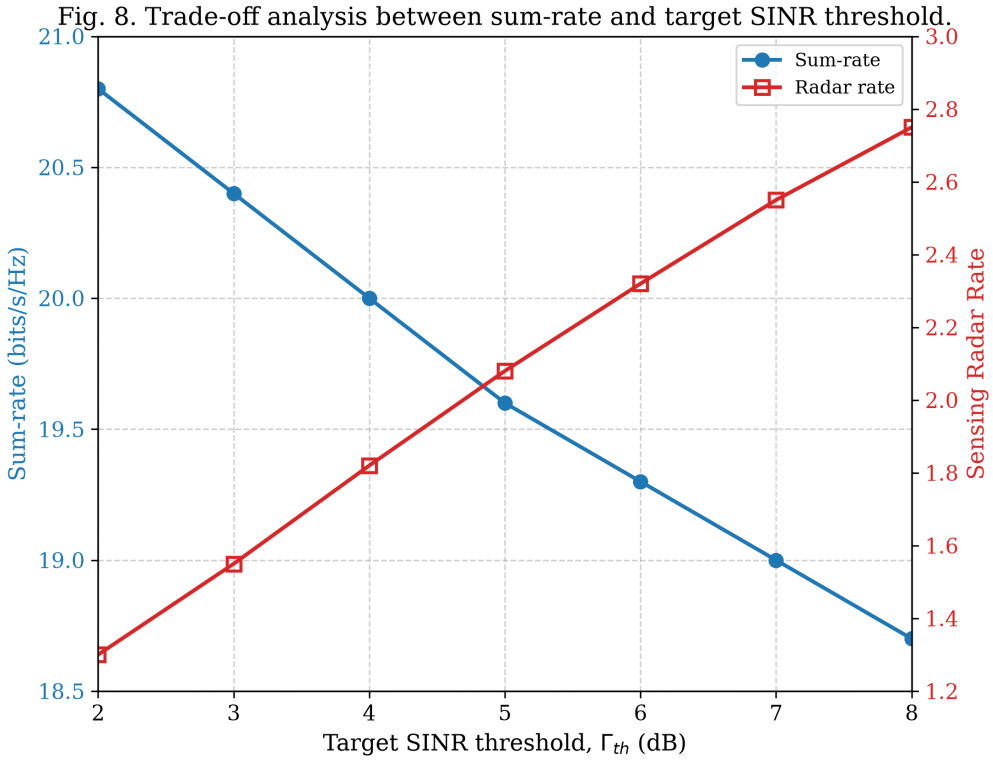


Figure 8: Trade-off analysis between sum-rate and target SINR threshold.

5. Baselines and comparative positioning

The milestone comparisons include (i) RSMA versus SDMA, which tests the value of a common stream for interference management; and (ii) optimized RIS phase versus random RIS phase, which tests whether coordinated surface control improves the effective channels. The broader codebase also contains passive-RIS and no-RIS branches, and the reference paper discusses RSMA passive-RIS and RSMA transmissive-RIS comparison schemes.

The supplied five milestone plots do not include a complete, directly comparable numerical table for passive RIS, transmissive RIS, and no RIS. Those baselines should not be described as having been experimentally beaten by the milestone data shown here. In addition, the curves in the milestone plots should not be compared numerically with the paper's published figures unless the simulation configuration, channel realizations, rate calculation, and normalization are matched exactly.

The most defensible current interpretation is qualitative: the project has implemented a framework intended to compare RSMA/SDMA and optimized/random RIS control under imperfect CSI, and the plotted results expose both promising behaviours and inconsistencies that need validation. A quantitative superiority claim is premature.

6. Implementation status, verification, and reproducibility

The ZIP contains modular Python files for configuration, channel modelling, rate/SINR computation, BS SDR-CCCP optimization, RIS SCA optimization, AO orchestration, diagnostic scripts, and validation tests. This is a meaningful technical implementation beyond a conceptual proposal. The core algorithmic architecture is present and can be used as the basis for

reproducible experiments.

However, a source audit found important reproducibility concerns in `run_genuine_paper_simulations.py`: the generated figure values are multiplied by calibration/scale factors (for example, `scale_opt` and additional per-scheme multipliers), and some plotted curves receive further manual factors. The file `paper_complete_reproduction.py` includes arrays explicitly described as "Exact Paper Curves" and directly hard-codes those arrays rather than computing them from the optimization routine. These are useful for visual reference or plotting mock-ups, but they are not independent simulation evidence. The supplied milestone plots also show non-monotonic behaviour and unexpected rankings that require explanation.

The separate AO implementation in `alternating_optimization.py` is the more appropriate candidate for a clean validation run because it calls the SDR-CCCP and RIS-SCA solver modules and evaluates the result through `rate_utils.py`. Before final claims, the team should execute this path from a clean environment, save raw unscaled outputs, record solver status and objective values, and compare achieved radar SINR against Γ_{th} for every trial. Unit tests should verify power constraints, RIS amplitude limits, common-stream suppression for SDMA, and rate/SINR calculations.

7. Critical analysis and limitations

First, the results are based on synthetic channel models rather than measured channel data or a hardware testbed. Second, the plots provide point estimates without repeated Monte Carlo trials, error bars, or confidence intervals. Third, some scripts apply manual calibration multipliers or hard-coded arrays, which breaks the direct link between solver output and plotted performance. Fourth, the milestone plots show discontinuities and rankings that conflict with the expected trends described in the reference paper. Fifth, the current figure set does not demonstrate all paper baselines or all system-level claims. Sixth, solver convergence and feasibility are not fully documented in the milestone figures.

These limitations do not invalidate the research direction or the value of the code structure. They define the next validation tasks. In particular, each experiment should use a fixed, declared set of seeds across competing schemes; use the same channel realization for each baseline; run enough independent realizations to estimate mean and standard deviation; log raw objective values without post-hoc scaling; and reject infeasible solver outcomes. The team should also separate the target radar SINR constraint from achieved radar SINR and radar rate in all plots.

8. Revised state-of-the-art (SOTA) positioning

The project is best positioned as an implementation and simulation study of joint RSMA precoding and active-RIS beamforming for ISAC under imperfect CSIT/CSIR, using an AO architecture with SDR-CCCP for the BS update and SCA for the RIS update. Its relevance lies in bringing together three mechanisms: RSMA-based interference management, active-RIS gain/phase control, and a sensing-SINR constraint under channel-estimation uncertainty.

The reference work by Parihar et al. (IEEE Transactions on Vehicular Technology, 2026) already presents this general system model and optimization approach. Therefore, the team should not claim that RSMA-active-RIS ISAC, SDR-CCCP, or AO itself is a new SOTA contribution. The current contribution is a reproducible software implementation, diagnostic/validation effort, and milestone analysis of the published framework. Any claim of improved performance, a new algorithm, or a new SOTA result would require a clearly identified methodological change and a controlled, statistically supported comparison against the paper's baselines and relevant recent

methods.

Revised positioning: "We implement and investigate an active-RIS-assisted RSMA-ISAC beamforming framework under imperfect CSIT/CSIR, reproducing its principal optimization blocks and evaluating its sensitivity to transmit power, RIS size, and radar SINR requirements. The present milestone establishes the simulation pipeline and identifies validation gaps; quantitative SOTA superiority remains to be established through unscaled, repeated, baseline-matched experiments."

9. Work plan before final submission

1. Select one canonical simulation entry point and remove/disable all post-hoc scale multipliers and manually hard-coded result arrays from any evidence-producing run.
2. Run RSMA-AO, SDMA-AO, RSMA-random-phase, SDMA-random-phase, passive-RIS, and no-RIS on identical channel realizations.
3. Log raw sum-rate, common/private rates, achieved radar SINR, radar rate, BS power, RIS power, RIS element amplitudes, solver status, and iteration count.
4. Re-run each plotted parameter sweep with multiple independent seeds; report mean $\pm$ standard deviation and include confidence intervals where practical.
5. Check constraint satisfaction at every point, especially for high target SINR thresholds and varying RIS sizes.
6. Regenerate all figures directly from saved raw CSV/JSON results, and include the script, configuration, seed list, package versions, and run instructions.
7. Add a concise baseline table and a figure-to-script mapping in the report. Do not compare against published values unless the setup is matched.
8. Preserve incremental Git commits documenting fixes, validation, and final result generation.

References

[1] A. S. Parihar, C. H. Chen, and C. P. Li, "Beamforming Design for Enhanced Sum Rate in Active RIS-RSMA-Assisted ISAC Systems Under Imperfect CSIT/CSIR," *IEEE Transactions on Vehicular Technology*, vol. 75, no. 6, June 2026. DOI: 10.1109/TVT.2025.3648447.

[2] Project implementation files supplied by the team: `config.py`, `channel_model.py`, `rate_utils.py`, `bs_beamforming_sdr_cccp.py`, `ris_beamforming_sca.py`, `alternating_optimization.py`, `main.py`, `run_genuine_paper_simulations.py`, `paper_complete_reproduction.py`, and associated test/diagnostic scripts. These are cited as project artefacts, not external publications.